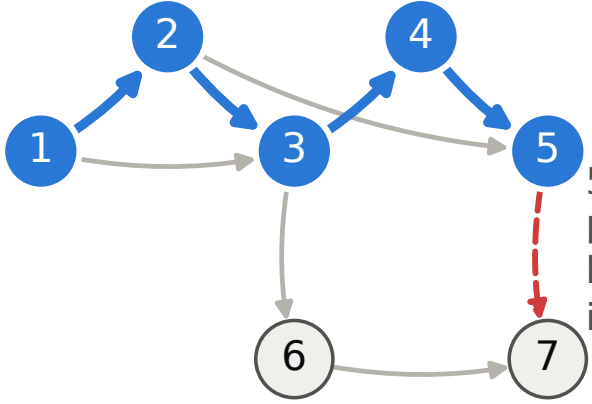
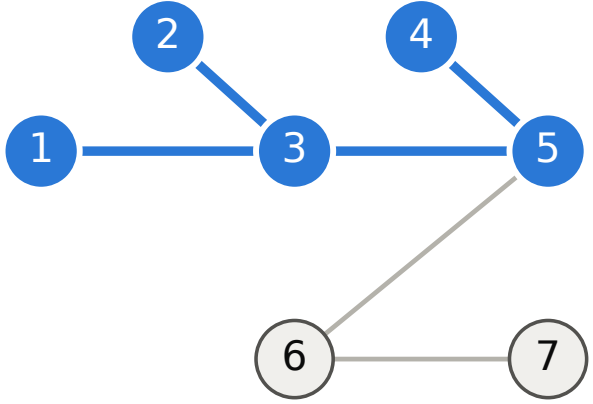
D (directed, acyclic): causal order



5→7 extends the path in *D*, but 7 has no *G*-neighbour in $\{1, \ldots, 5\}$

G (undirected): functional association

1 and 2 are not adjacent in *G*, yet $G[\{1, \ldots, 5\}]$ is connected